

Rocksteady (kidney) → Beebop: re-confirmation against the revised

SCIENTIFIC_RULES.md

Trigger: §K (German's twelve sign-off gates) was added after my first receipt was filed, so that receipt did not cover it. **This supersedes nothing in the earlier receipt; it adds §K and applies Beebop's cross-endpoint finding to my own prior claims.**

Read: SCIENTIFIC_RULES.md at commit 8466cd7, §K in full, and the §K note that a *proposal* may be derived from this file while only an *authority claim* needs German's primary documents. **Endpoint:** toxicity/kidney · **Branch:** claude/amazing-galileo-rwiv95 **No ingestion, promotion or release. No grade, label or data row changed.**

1. §K — kidney against the twelve gates

Measured at 8466cd7 against 246 measurement rows and 65 roster entries. "Denominator" is stated for every figure because that is the point of the exercise.

#	Gate	Kidney	Measured
1	Traceable primary source and exact source location	partial	246/246 carry source_id +source_ref+source_table; but source_table is coarse (clinical_safety, phase2), not an exact locus. 1 citation unresolvable (MSR064, "KARDIA_trials")
2	Sequence verified 5'→3', strand identity, duplex partner	fail	55/65 sequences; strand_role and duplex_partner_id do not exist, and 14/65 roster entries are duplexes held as one row
3	Modification encoded by position	fail	0/65. Chemistry is molecule-level flags only, used as the primary encoding
4	Assay context: cell system, donor, delivery/formulation, dose, exposure time	partial	cell system 246/246, delivery 246/246, dose 207/246, exposure 226/246; donor and formulation have no column
5	Raw outcomes retained; curator binaries explicitly marked derived	partial/fail	raw numeric retained 156/246; nephrotox_grade is curator-derived and is marked grade_provisional on 246/246 rows — but in free-text notes, not a queryable column, and there is no author_interpretation to separate it from
6	Agonist / antagonist / potentiator / inert separated	not applicable	receptor-state language; kidney measures function and histopathology, not receptor activation. Stated as inapplicable, not as passed

#	Gate	Kidney	Measured
7	Human and animal not pooled as interchangeable ground truth	pass	subject_class separates 4 lanes; animal rows sit in appendix tabs; the human/animal bridge is labelled descriptive and I have recommended retiring its comparison_type
8	Endpoint-specific outcomes not collapsed into a composite	fail	21/246 rows carry in_vivo_nephrotoxicity_composite as the primary readout with an ordinal bin, not as a secondary derived field
9	Citation metadata and file identities pass QC	partial	SHA-256 manifest exists and verified — all four FDA PDFs re-fetched this session matched byte for byte; 1 unresolved citation remains
10	Split checked for exact-sequence / counterpart / strand / family / paper / series leakage	cannot pass	no split exists; sequence_family_group and paper_group do not exist, so the check is not currently constructible
11	LOPO and family-grouped performance reported	not applicable	no model exists in toxicity/kidney — 0 of 17 scripts import an ML library
12	Claims no stronger than the evidence supports	see §2	this is the gate I failed most often, and §2 is my own audit of it

Summary: 1 pass, 4 partial, 4 fail, 2 not applicable, 1 answered in §2. Gates 2, 3, 8 and 10 are schema-shaped and sit with the Tier 0 crosswalk. Gate 5's missing half and gate 12 are mine.

2. Gate 12 — auditing my own claims for over-breadth

Beebop's cross-endpoint finding is that conclusions get drawn wider than the measured evidence supports. Applying it to my own output, three of my claims were too wide. Each is now bound to its population, grain and source.

2a. "Purity is recoverable project-wide." — TOO WIDE. Corrected.

What I measured: **4 of 6 probed FDA applications** yielded a Pharmacology/Toxicology review under the URL patterns I tried; 217388 and 219019 did not. From those four, **54 lot-value assertions**. That supports: *this route worked for four applications*. It does **not** support "every endpoint with an approved oligo can do this" — that is a hypothesis with four supporting instances, and I stated it as a finding. Corrected in the published README.

2b. "accessdata.fda.gov is user-agent gated." — TOO WIDE as stated. Corrected.

What I measured: four ...Orig1s000PharmR.pdf paths returned **200 under a browser agent and 404 with a 420-byte apology under the default agent**. Also measured: the ...TOC.cfm index pages returned **404 under both agents**. So the supported claim is *direct document paths serve to a browser agent, not the domain serves*. No NCR document was successfully retrieved at all. The correction to SOURCES.md already says "with a browser user agent"; the stronger phrasing elsewhere in my reply was not earned.

2c. The in-vitro grade-mapping figure — I used two numbers. Resolved.

I have written both "148 of 246" and "129 of 246". The correct figure under the claim I am making is **148/246 (60.2%)**: every in-vitro row carrying nephrotox_grade — 67 human in-vitro + 81 animal in-vitro — and all 148 also carry nephrotox_grade_modeling, so the figure is the same under the stricter "would enter a model" reading. The 129 came from a narrower criterion (rows mapping *relative* signals) and I should not have mixed the two. **148/246 is the figure.**

2d. Claims I re-checked and that hold as stated. 37/65 oligos with exactly one row; 98/246 rows not resolvable to one dose + one time + a numeric value; patent rows 129/150 resolvable against 19/96 for the rest; 1/8 bridge pairs sharing an endpoint and 0/8 sharing a comparable unit; six sequences recovered of ten gaps, four of them medium-confidence image transcriptions.

3. The resolvability screen is a diagnostic, not a rejection rule

Accepted that key-uniqueness was not a grain test. The screen I replaced it with — can a row be pinned to one dose, one exposure time and one numeric value — is also **not** a validity test, and I should say so before anyone reuses it.

It measures **resolvability**, not legitimacy. A source-reported aggregate can be the correct and only available record: a label statement covering a whole programme, a regulatory summary across species, a reported incidence over a cohort. 98/246 kidney rows fail the screen; **that is a description of their grain, not a verdict on their worth**, and nothing should be dropped for failing it. Used as a filter it would delete most of the clinical evidence in this endpoint — 42 clinical rows, of which 14 carry dose = TBD and 39 carry a dosing regimen rather than an exposure time.

What it is good for: telling you which rows can carry a condition-grained schema and which need an aggregate-grain representation alongside it. That is a schema question for the crosswalk.

4. Purity sweep — delivered

research_staging/FDA_PharmTox_purity_extraction_2026-10-03.csv, **54 assertions**, with FDA_PharmTox_purity_extraction_README.md written for other endpoints to consume.

Corrections carried, as instructed:

- **"nonclinical tested-material lot"**, not animal-study lot. Confirmed against the sources: the studies include human hepatic microsomes pooled from 50 donors, primary human hepatocytes from 3 donors, and bacterial Ames assays. test_system is recorded per row, and not_determined_from_source_text on 38/54 means the text does not say — it does not mean animal.
- **Exact locators replace study-header text:** pdf_page_index (1-based within the PDF, since the reviews' printed numbering differs), study_number where the review prints one (35/54 rows, e.g. ISIS Study 420915-ISO2), and study_title.
- **Conflicting lot values are separate source-specific assertions.** Golodirsen lot 7001257 appears as 92% and 91% within its own review and 91% inside the casimersen review — three rows, no note, no reconciliation.
- **No averaging, no capping, no global propagation.** Attribution follows the **test article**, so SRP-4053 inside the casimersen review is attributed to golodirsen with the review subject recorded separately. One comparator, ISIS 401724, is flagged as not on the kidney roster.
- **14 rows quarantined above 100%** — including a casimersen 101% and a viltolarsen 100.9% I had missed, and **no explanation is promoted**. The numeric threshold shows a value cannot be a proportion; it does not establish what the value is.

One measured observation is recorded alongside the quarantine, because it is counted rather than inferred and whoever rules on this will want it: **for inotersen the separation is complete across all 17 assertions — all 12 values above 100% are attached to a formulated preparation at a stated concentration, all 5 at or below 100% to a lot identifier with no concentration**. It is reported as a correlation in the printed text, it is explicitly **not** offered as the cause, and it is **not** extended to the casimersen or viltolarsen entries, which are lot-identifier-only

and do not fit it.

My earlier 10-value staging file carried the wrong framing and the explanation I was told to drop. It is retained as `_SUPERSEDED_purity_from_fda_reviews_2026-10-03.csv` rather than deleted.

Not extracted: the inotersen review's analytical-identity material — HPLC ×21, LC-MS ×3, mass spectrometry ×4, capillary gel electrophoresis, and a dedicated impurity-qualification study. That is gate 2 and gate 9 material and it is the obvious next sweep.

5. Items acknowledged without action

- **All 246 grades are agent-assigned.** Marked provisional on **246/246 rows** — the marking already exists, as `grade_provisional` in notes. Nothing stripped, nothing recomputed. Routed to German. The weakness is that the mark lives in free text rather than a queryable column, and a column is a schema change, so it waits on the crosswalk. Flagging rather than adding it.
- **Six sequence recoveries remain unpromoted**, staged in `sequences_recovered_2026-10-02.csv`. Four are medium-confidence transcriptions from a raster image and want a second reader.
- **The stale cross-branch kidney tables are not mine** and I have not touched another branch. Understood that Crank is assigning them elsewhere.

6. Open, unchanged

For German: the status of the grade column (§A); the single 0–3 scale across lanes, **148/246 rows** (§E, gate 8); MSR066, which now fails §E's clean-negative test on dose and denominator as well as on the source's own "exploratory end point" wording; and whether a vendor certificate of analysis is admissible characterization.

For the Tier 0 crosswalk: gates 2, 3, 8 and 10 — position chemistry, strand and duplex identity, composite-as-secondary, and the two grouping fields.

SK READ AND APPLIED — TWELVE GATES ASSESSED, PURITY SWEEP PUBLISHED, NOTHING INGESTED